

Evolution of dietary antioxidants.

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Oxygen is vital for most organisms but, paradoxically, damages key biological sites. Oxygenic threat is met by antioxidants that evolved in parallel with our oxygenic atmosphere. Plants employ antioxidants to defend their structures against reactive oxygen species (ROS; oxidants) produced during photosynthesis. The human body is exposed to these same oxidants, and we have also evolved an effective antioxidant system. However, this is not infallible. ROS breach defences, oxidative damage ensues, accumulates with age, and causes a variety of pathological changes. Plant-based, antioxidant-rich foods traditionally formed the major part of the human diet, and plant-based dietary antioxidants are hypothesized to have an important role in maintaining human health. This hypothesis is logical in evolutionary terms, especially when we consider the relatively hypoxic environment in which humans may have evolved. In this paper, the human diet is discussed briefly in terms of its evolutionary development, different strategies of antioxidant defence are outlined, and evolution of dietary antioxidants is discussed from the perspectives of plant need and our current dietary requirements. Finally, possibilities in regard to dietary antioxidants, evolution, and human health are presented, and an evolutionary cost-benefit analysis is presented in relation to why we lost the ability to make ascorbic acid (vitamin C) although we retained an absolute requirement for it.